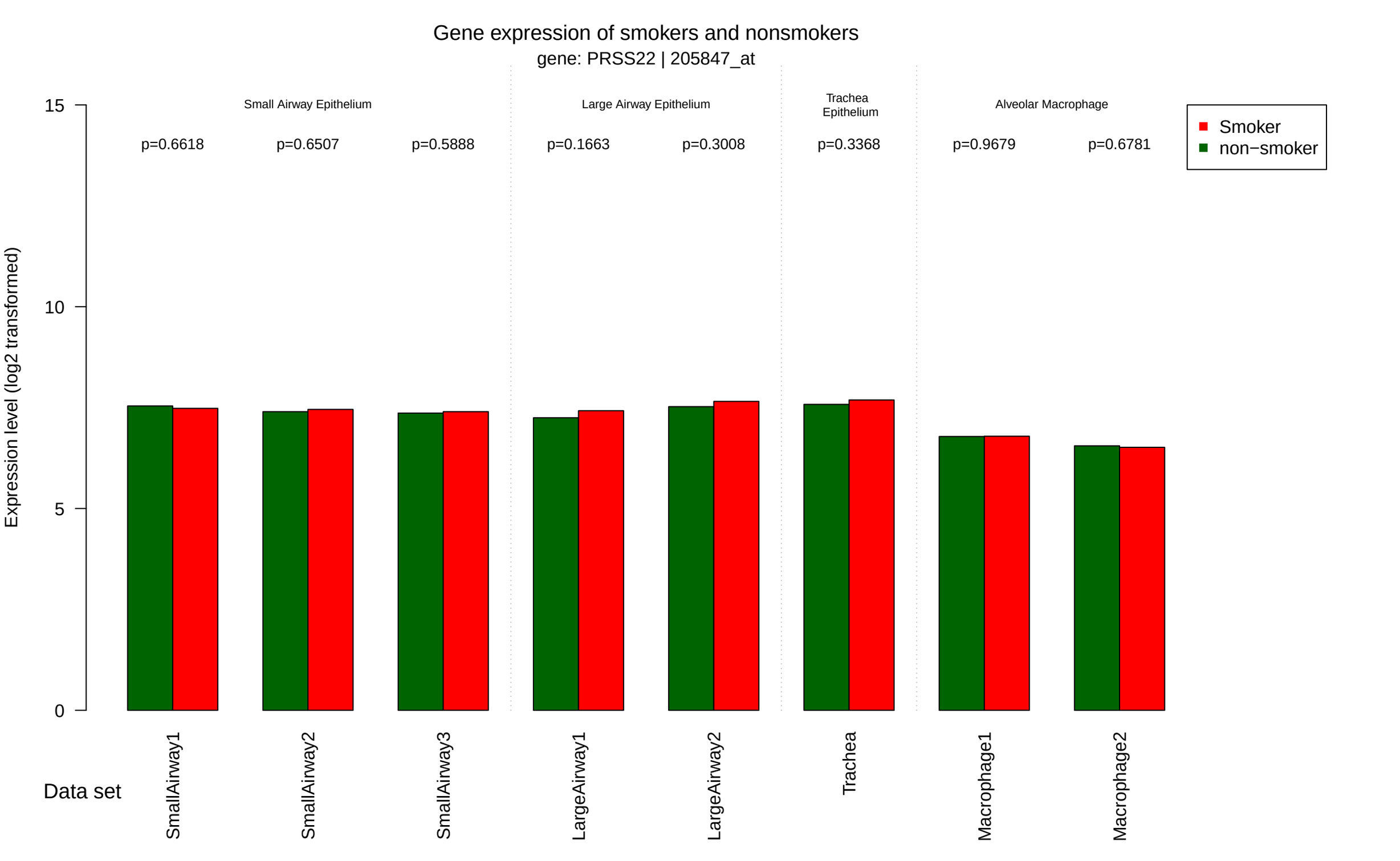
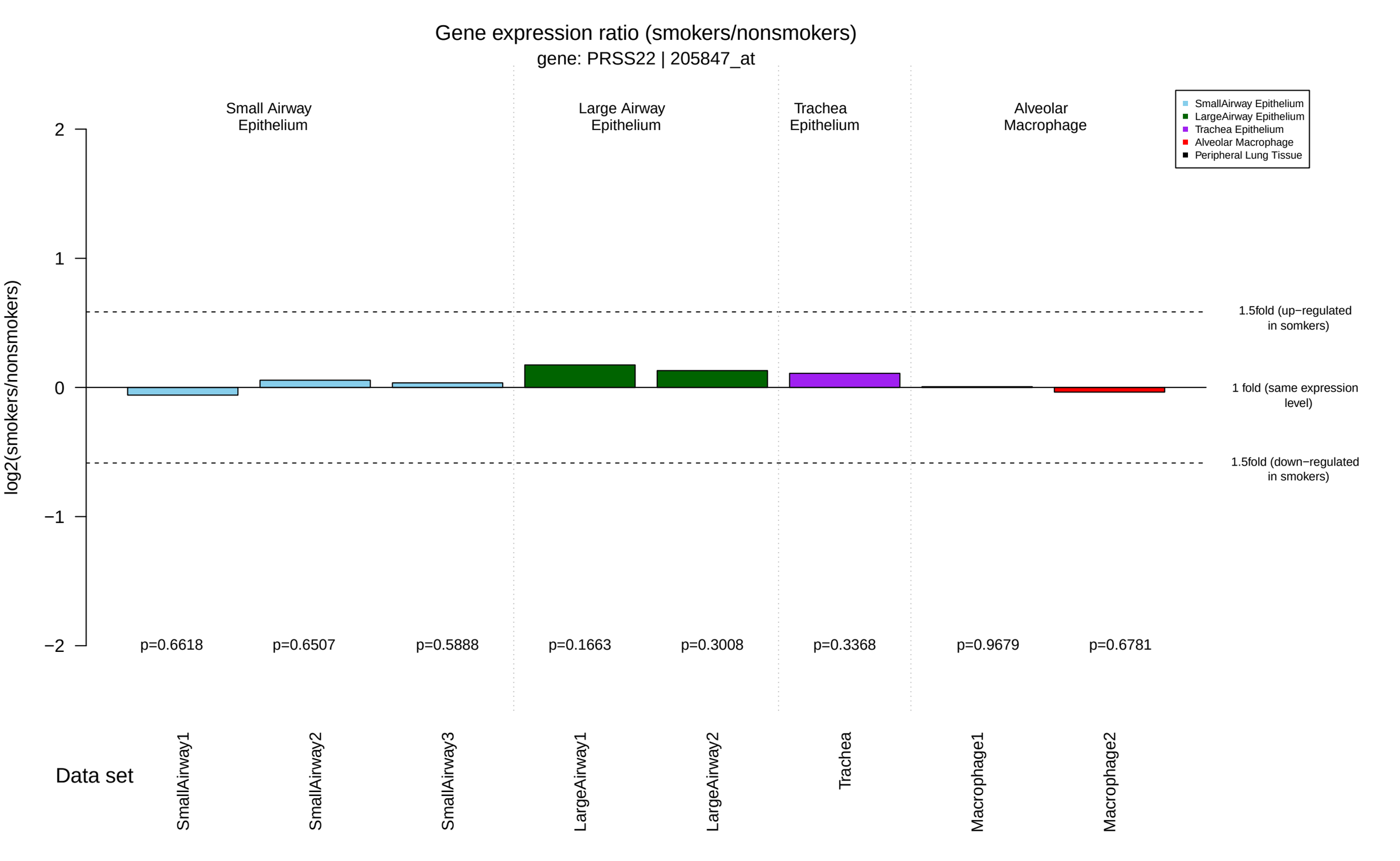
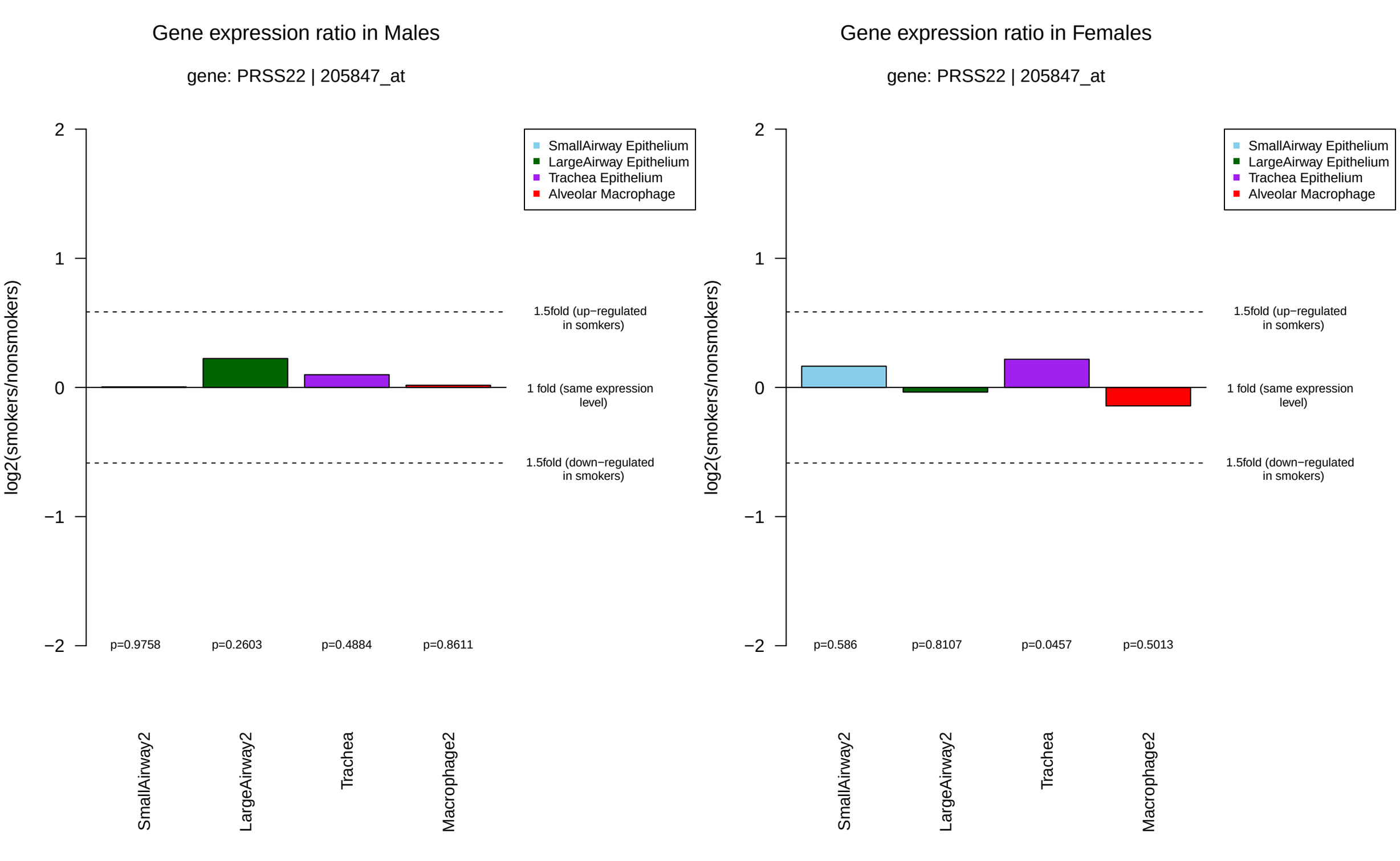
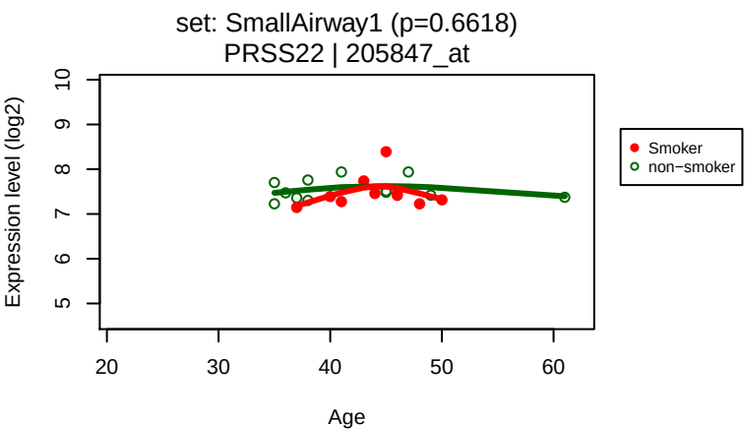
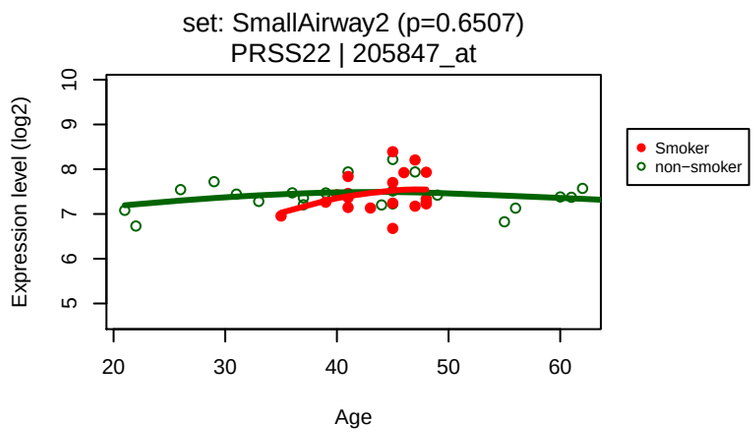
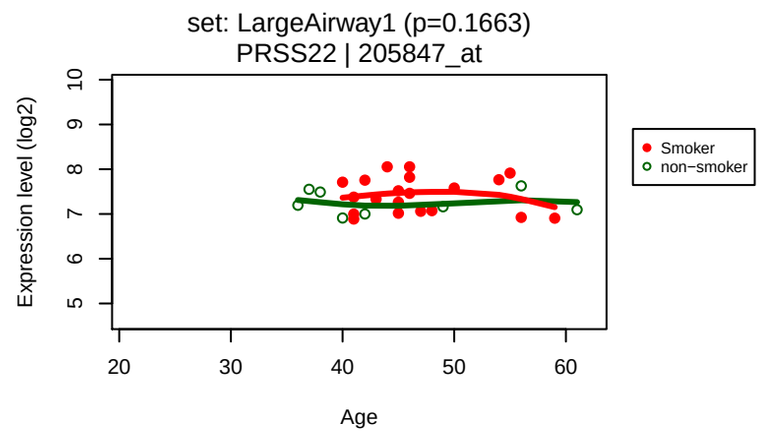

Gene expression of smokers and nonsmokers
gene: PRSS22 | 205847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6618
p=0.6507
p=0.5888
p=0.1663
p=0.3008
p=0.3368
p=0.9679
p=0.6781
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

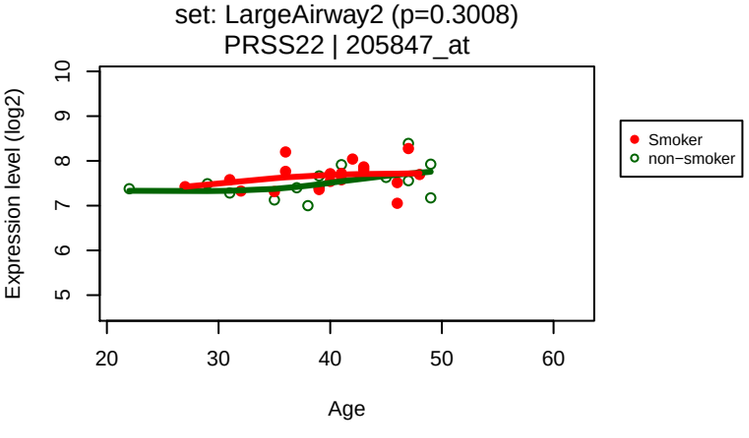

Gene expression ratio (smokers/nonsmokers)
gene: PRSS22 | 205847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6618
p=0.6507
p=0.5888
p=0.1663
p=0.3008
p=0.3368
p=0.9679
p=0.6781
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

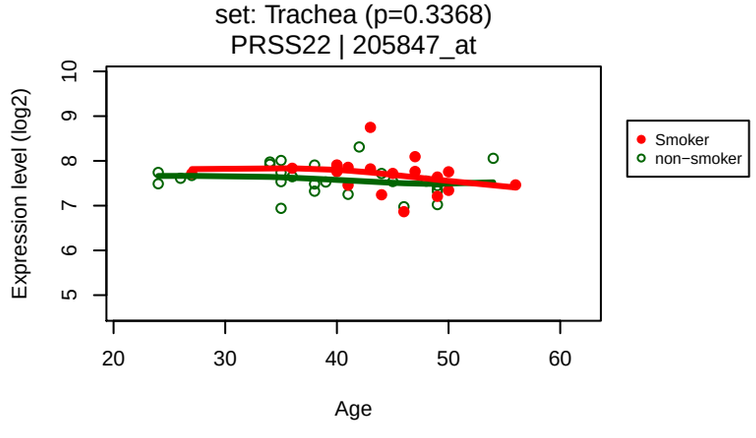

Gene expression ratio in Males
gene: PRSS22 | 205847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9758
p=0.2603
p=0.4884
p=0.8611
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS22 | 205847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.586
p=0.8107
p=0.0457
p=0.5013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

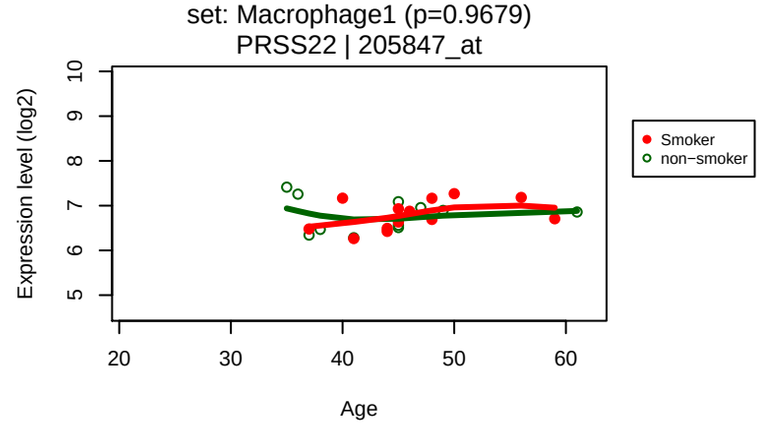

set: SmallAirway1 (p=0.6618)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker

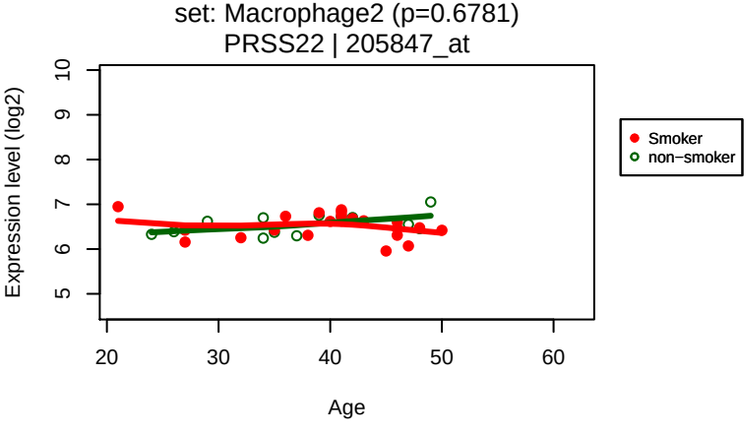

set: SmallAirway2 (p=0.6507)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker

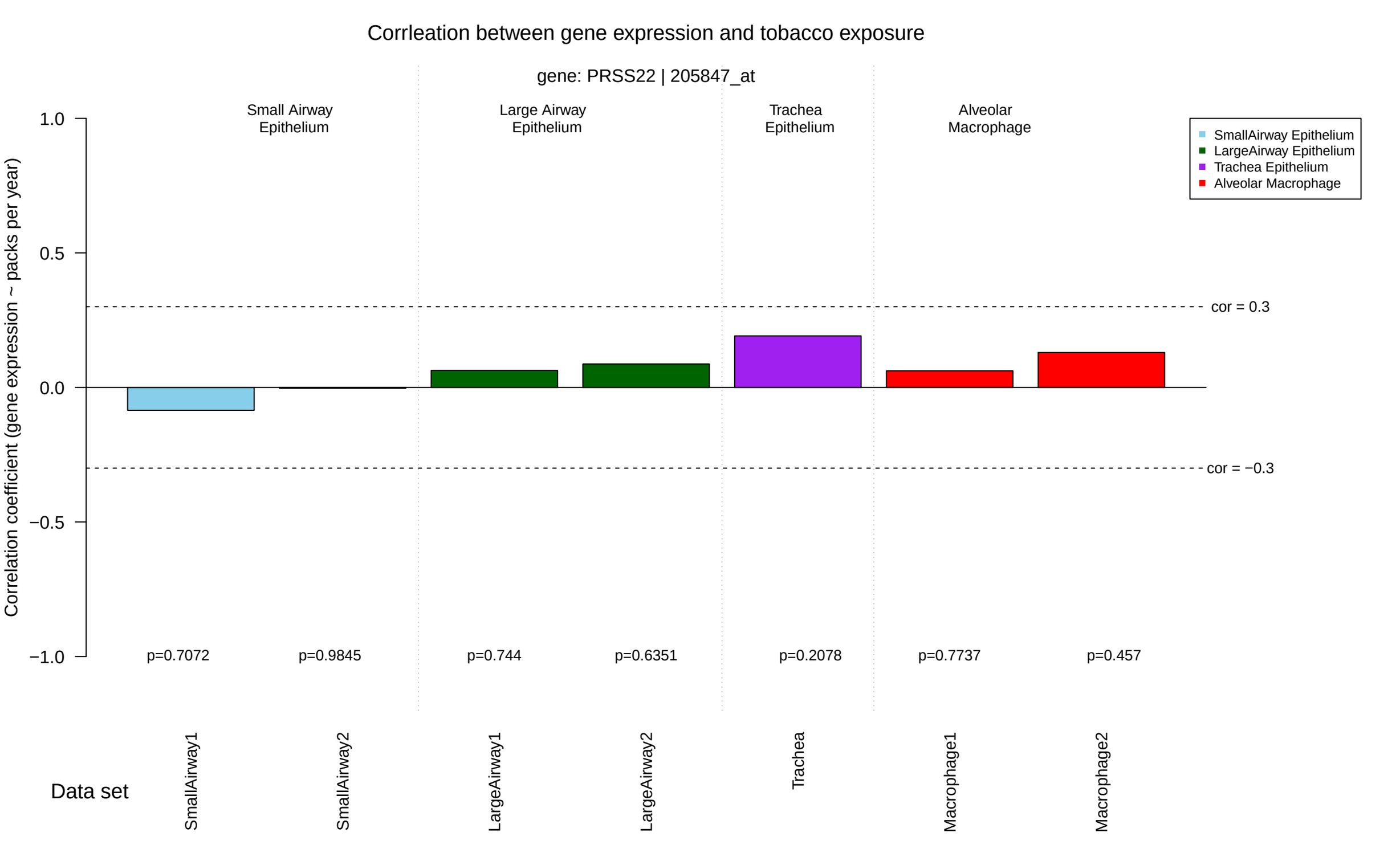

set: LargeAirway1 (p=0.1663)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3008)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3368)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9679)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6781)
PRSS22 | 205847_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS22 | 205847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7072
p=0.9845
p=0.744
p=0.6351
p=0.2078
p=0.7737
p=0.457
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2